

Gregory Eugene Jones



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PROFESSIONAL EXPERIENCE |

Senior ML Engineer, Natera — Remote

Nov 2023 — Present

- Developed and deployed advanced machine learning models using TensorFlow, Scikit-Learn, and PyTorch, leveraging techniques like regression, Random Forests, PCA, clustering (K-Means, DBSCAN), NLP (NER, tokenization), and outlier detection to extract actionable insights from complex healthcare data.
- Designed scalable and efficient data pipelines using Apache Spark and Databricks, processing over 2TB of data monthly with minimal latency, enabling seamless data transformation and model training at scale.
- Built and optimized Retrieval Augmented Generation (RAG) pipelines using Weaviate and LangChain to enhance enterprise knowledge bases, achieving a 93% query accuracy for large-scale document retrieval.
- Fine-tuned large language models (LLMs) using Hugging Face Transformers, incorporating domain-specific data and custom training pipelines to achieve enhanced performance in specialized healthcare applications.
- Engineered cloud-based solutions on AWS, utilizing services such as S3, EC2, Lambda, and SageMaker to deploy scalable and reliable machine learning workflows.
- Architected enterprise ETL pipelines to extract and process data from data lakes, ERPs, and MemSQL clusters, integrating REST APIs, Airflow, and Dynamic SQL for seamless updates to data warehouses.
- Developed APIs and services using Flask for deploying machine learning models, enabling real-time predictive capabilities and seamless integration with the Constellation Platform.
- Implemented AutoML frameworks for neural architecture search, hyperparameter optimization, and multi-domain learning, improving model robustness and reducing experimentation time.
- Integrated traditional and generative AI models into WhyLabs via CI/CD pipelines, enhancing monitoring, governance, and performance metrics across the security ecosystem.
- Specialized in deep learning techniques such as knowledge distillation, transfer learning, and multimodal learning for improving healthcare data insights and predictive accuracy.
- Created comprehensive quality assurance (QA) suites in Python, performing regression, integration, and unit testing while ensuring data integrity with custom QC checks and Type 2 error tracking.
- Visualized complex data insights for both technical and non-technical stakeholders using Matplotlib, Seaborn, and Plotly, presenting compelling narratives to drive data-driven decisions.
- Collaborated with healthcare professionals to integrate domain expertise into predictive models, improving accuracy in cell-free DNA testing.
- Designed model monitoring dashboards using Grafana and Prometheus, enabling real-time tracking of model performance and ensuring operational stability.
- Explored MLOps strategies by integrating Kubeflow pipelines for end-to-end model lifecycle management, enabling reproducibility and scalability in deployment.

ML Engineer, Onspring — Remote

Jun 2021 — Oct 2023

- Implemented an LLM-based RAG system using LlamaIndex for advanced product analytics by integrating Google Analytics API data, which enhanced data-driven decision-making and resulted in a 25% increase in user engagement and a 20% improvement in content recommendation accuracy.
- Developed and deployed AI-driven chatbot systems using Amazon Bedrock, incorporating API endpoints and voice-to-chat transcription pipelines powered by Amazon Transcribe, which enhanced client interactions and enabled seamless conversational AI automation.
- Integrated a machine learning forecasting model using Vector Error Correction Models (VECM), achieving a Mean Absolute Percentage Error (MAPE) of 13% to deliver accurate predictions and actionable insights for

informed decision-making.

- Leveraged microservice integration and deployment techniques alongside an inference manager, achieving a 15% improvement in recognition accuracy and a 20% reduction in processing time.
- Created a POC for OctoAI's cloud generative AI platform, showcasing text generation with large language models and image generation with stable diffusion, while evaluating the Asset Orchestrator for fine-tuning images using custom weights from a limited set of subject images.
- Performed extensive data preprocessing and quality assurance using Pandas and NumPy, ensuring data consistency, integrity, and readiness for analysis and model training.

AI Software Engineer, Bloomreach — Mountain View, CA

Mar 2015 — May 2021

- Developed, productionized, and deployed advanced machine learning models for recommender systems, leveraging techniques like matrix factorization and neural network architectures (RNNs, LSTMs, AutoEncoders) using Python, TensorFlow, and XGBoost, with deployment on iOS via CoreML and computation on Microsoft Azure, resulting in measurable improvements in user experience and content discoverability.
- Implemented Bayesian optimization and online learning techniques to enhance ML model performance and scalability in real-time environments.
- Built and improved data pipelines for machine learning workflows using Apache Spark and SQL, enabling efficient model training and real-time data processing.
- Conducted large-scale A/B testing and online experimentation leveraging hypothesis testing, multi-armed bandit algorithms, and causal inference techniques to validate and refine model performance in production environments.

Data Analyst, Amazon — Seattle, WA

Mar 2013 — Feb 2015

- Analyzed chatbot data on 12 custom criteria, resulting in improved accuracy and data quality.
- Designed and engineered 100+ prompts to evaluate Amazon AI models leading to performance optimization.
- Improved AI model output diversity by 40% with the help of 5 novel data augmentation strategies.
- Conducted extensive model performance evaluations and testing, refining algorithms to achieve high precision and recall rates.

EDUCATION |

North Carolina State University, Raleigh, NC

Oct 2012

Bachelor of Science in Data Science: GPA: 3.82

Bachelor of Science in Computer Science: GPA: 3.82

- Relevant Coursework: Data Structures and Analysis, Probability, Statistical Theory, Data Mining and Machine Learning, Distributed Computer Architecture, Object Oriented and Systems Programming with Python, Linear Algebra, and Multivariable Calculus

SKILLS |

- **Languages**: Python, R, SQL, Java, C/C++, Bash, HTML, CSS, Javascript
- **Tools**: Git, Linux, AWS, SageMaker, MongoDB, MLOps, Power BI, Tableau
- **Technical Proficiency**: Machine Learning, Deep Learning, PyTorch, Large Language Models (LLM), Hugging Face, Natural Language Processing (NLP), Transformer Models
- **Soft Skills**: Adaptability, Attention to detail, Work Ethic, Problem-Solving, Leadership